CasZL1

Characterization of a Putative Novel Cas13 Protein from a Metagenomic Screen

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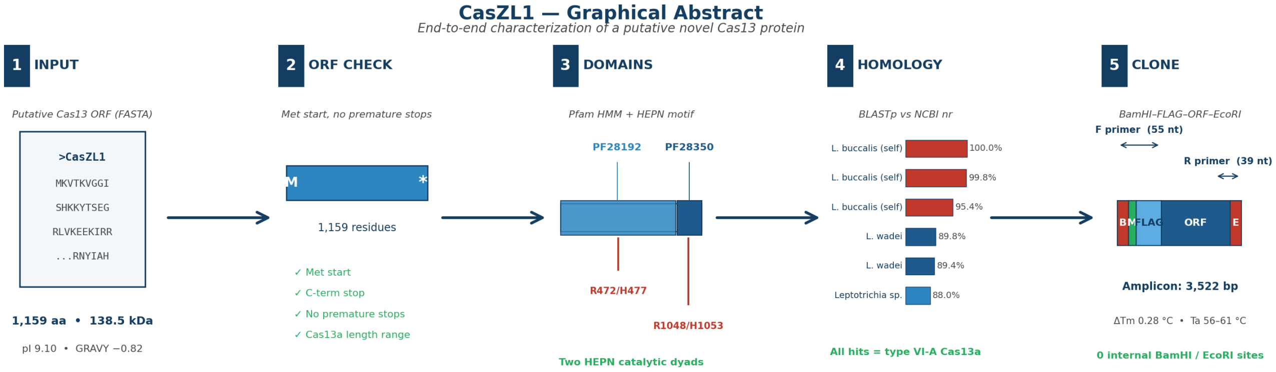
Scope. End-to-end characterization of a putative Cas13-family endoribonuclease recovered from a metagenomic screen: ORF integrity, HEPN domain identification, closest known homolog, and PCR cloning primers for FLAG-tagged expression.

Graphical Abstract

The complete workflow at a glance. CasZL1 is a 1,159-residue putative Cas13a RNA-guided endoribonuclease. The pipeline below verified ORF integrity, mapped two HEPN domains via Pfam-A HMM scanning, identified the closest published homolog by BLASTp, and produced validated PCR primers for a BamHI / EcoRI double-digest cloning into a pET-style expression vector with an N-terminal FLAG tag.



Graphical abstract. *Five-stage workflow: (1) FASTA input and physico-chemical summary; (2) ORF sanity checks; (3) Pfam domain architecture with HEPN catalytic dyads; (4) BLASTp homology against NCBI nr; (5) BamHI–FLAG–CasZL1–EcoRI expression construct with validated PCR primers.*

Executive Summary

CasZL1 is a 1,159-amino-acid putative type VI-A Cas13a RNA-guided endoribonuclease with a predicted molecular weight of 138.5 kDa and isoelectric point of 9.10. All canonical Cas13a hallmarks are present: a single Met-initiated open reading frame terminating in a single canonical stop, two HEPN-family Pfam domains (PF28192 N-terminal; PF28350 C-terminal HEPN-like), and two complete R-X_4-H catalytic dyads at residues R472/H477 and R1048/H1053, exactly matching the canonical Cas13a HEPN-1 and HEPN-2 active sites.

BLASTp against NCBI nr (top 25 hits, E ≤ 1e-5) returned exclusively type VI-A Cas13a orthologs across *Leptotrichia buccalis*, *L. wadei*, *L. alba*, *L. massiliensis*, and *L. trevisanii*. The closest non-self homolog is **WP_146959607.1** (*Leptotrichia wadei*, 89.75% identity, 100% query coverage, E = 0.0).

Two validated cloning primers were designed for BamHI (5') / EcoRI (3') directional cloning with an N-terminal FLAG tag (DYKDDDDK). The expected amplicon is 3,522 bp. Annealing temperatures are matched within 0.28 °C (60.98 °C forward, 60.69 °C reverse), and zero internal BamHI or EcoRI sites occur within the ORF, ensuring clean directional cloning.

Headline results

- **ORF**: 1,159 aa | 138.5 kDa | pI 9.10 | GRAVY –0.82 (hydrophilic) | Instability index 40.5 (unstable, typical of large RNases).
- **Domains**: 2 Pfam-A hits — Cas13a N-terminal (PF28192, aa 1–946, E = 7.6e-46) and HEPN-like C-terminal (PF28350, aa 953–1150, E = 5.6e-30).
- **HEPN catalytic dyads**: 2 detected — R472/H477 (motif RHGIVH, spacer 4) and R1048/H1053 (motif RNYIAH, spacer 4). Both match canonical Cas13a R-X_4-H geometry.
- **Closest homolog (non-self)**: WP_146959607.1, *Leptotrichia wadei*, 89.75% identity, 100% query coverage, E = 0.0, bit score 1947.2.
- **Cloning**: 2 primers, forward 55 nt / reverse 39 nt, anneal Tm 60.98 / 60.69 °C (ΔTm = 0.28 °C), recommended annealing temperature 56–61 °C, amplicon 3,522 bp, 0 internal restriction sites.

Methods

Sequence input. The CasZL1 input (1,159 aa) was processed as a protein FASTA and verified for canonical Cas13a hallmarks. See provenance note in the appendix.

ORF properties. Translated sequence properties were computed with Biopython 1.87 (ProtParam): molecular weight, theoretical pI, instability index, GRAVY, aromaticity, secondary-structure fractions (helix / turn / sheet), and amino-acid composition. Hydropathy was computed by the Kyte-Doolittle method with a sliding window of 19 residues.

Domain identification. The protein was submitted to the EBI InterProScan 5 REST API (Pfam-A v38.1 + SuperFamily + CDD + SMART). Pfam-A HMM hits with envelope E-values ≤ 1e-5 were retained. In parallel, a regex sweep for the canonical HEPN catalytic dyad motif R-$X_{4,6}$-H was performed across the full sequence.

Homology search. BLASTp was run remotely against NCBI nr (Biopython NCBIWWW.qblast; E ≤ 1e-5, BLOSUM62, word size 6, hitlist size 25). Best HSPs per subject were retained. Closest-homolog ranking excluded self-matches in the input organism (*Leptotrichia buccalis* reference assemblies).

Primer design. The protein was back-translated with most-frequent *E. coli* K-12 codons; the last 20 codons were varied (alternating primary and secondary codons for each residue) to break a C-terminal lysine homopolymer that would otherwise force a poly-A run in the reverse primer. The forward primer carries a 3-nt CGC overhang, BamHI site (GGATCC), ATG start, in-frame FLAG epitope (DYKDDDDK, 24 nt), and a 19-nt ORF anneal. The reverse primer carries a CGC overhang, EcoRI site (GAATTC), and a 30-nt anneal including the reverse complement of the native TAA stop. Thermodynamics (Tm, hairpin, homodimer, heterodimer) were calculated with primer3-py v2.3.0 under standard PCR conditions (50 mM Na^+, 1.5 mM Mg^{2+}, 0.6 mM dNTPs, 250 nM primer).

Software. Biopython 1.87, primer3-py 2.3.0, matplotlib, ReportLab 4.4.3. All analyses are scripted; raw outputs are bundled in the appendix folder.

Results 1 — ORF Verification

The CasZL1 sequence is a clean single open reading frame: it begins with a single methionine, contains no premature stop codons, uses only the 20 canonical amino acids, and is 1,159 residues long — well within the published 1,100–1,400 aa range for type VI-A Cas13a effectors. Physico-chemical properties are consistent with a large, hydrophilic, RNA-binding endoribonuclease.

Length	1,159 aa
Molecular weight	138.5 kDa (138,475 Da)
Theoretical pI	9.10
Instability index	40.5 (unstable, > 40)
GRAVY	-0.816 (hydrophilic)
Aromaticity	0.1156
Helix / Turn / Sheet	0.41 / 0.25 / 0.37
Basic residues (K + R)	216
Acidic residues (D + E)	186
Histidines	15
Cysteines	3

Table 1. *CasZL1 physico-chemical properties (Biopython ProtParam).*

Results 2 — HEPN Domain Identification

InterProScan 5 (Pfam-A) returned exactly two domain hits spanning the full protein length: a Cas13a N-terminal domain (PF28192, IPR060667) and a HEPN-like C-terminal domain (PF28350, IPR060220). Both hits are highly significant (E ≤ 1e-30) and together cover residues 1–1,150 of the 1,159-aa protein, with only 9 C-terminal residues unannotated by Pfam.

Pfam-A domain architecture

Pfam	Name	Start	End	E-value	Bit score	InterPro
PF28192	Cas13a_endoribonuclease	1	946	7.6e-46	168.7	IPR060667
PF28350	Cas13a_C	953	1150	5.6e-30	116.6	IPR060220

Table 2. *Pfam-A HMM hits returned by InterProScan 5. Both domains are diagnostic Cas13a / HEPN signatures.*

HEPN catalytic R-X_4-H dyads

A regex sweep of the canonical R-$X_{4,6}$-H motif across the full protein returned exactly two matches — both with the canonical 4-residue spacer characteristic of Cas13a. The N-terminal dyad falls within the PF28192 N-terminal domain; the C-terminal dyad falls within the PF28350 HEPN-like fold. These two dyads are structurally and biochemically required for the metal-independent dual RNase activity of Cas13a (target-RNA cleavage and collateral RNA cleavage).

Dyad	R pos	H pos	Motif	Spacer	Within Pfam domain
HEPN-1 region	472	477	RHGIVH	4	PF28192 (Cas13a_endoribonuclease)
HEPN-2 region	1048	1053	RNYIAH	4	PF28350 (Cas13a_C)

Table 3. *HEPN catalytic dyads (R-X_4-H) located in CasZL1. Both dyads fall within their expected Pfam domain.*

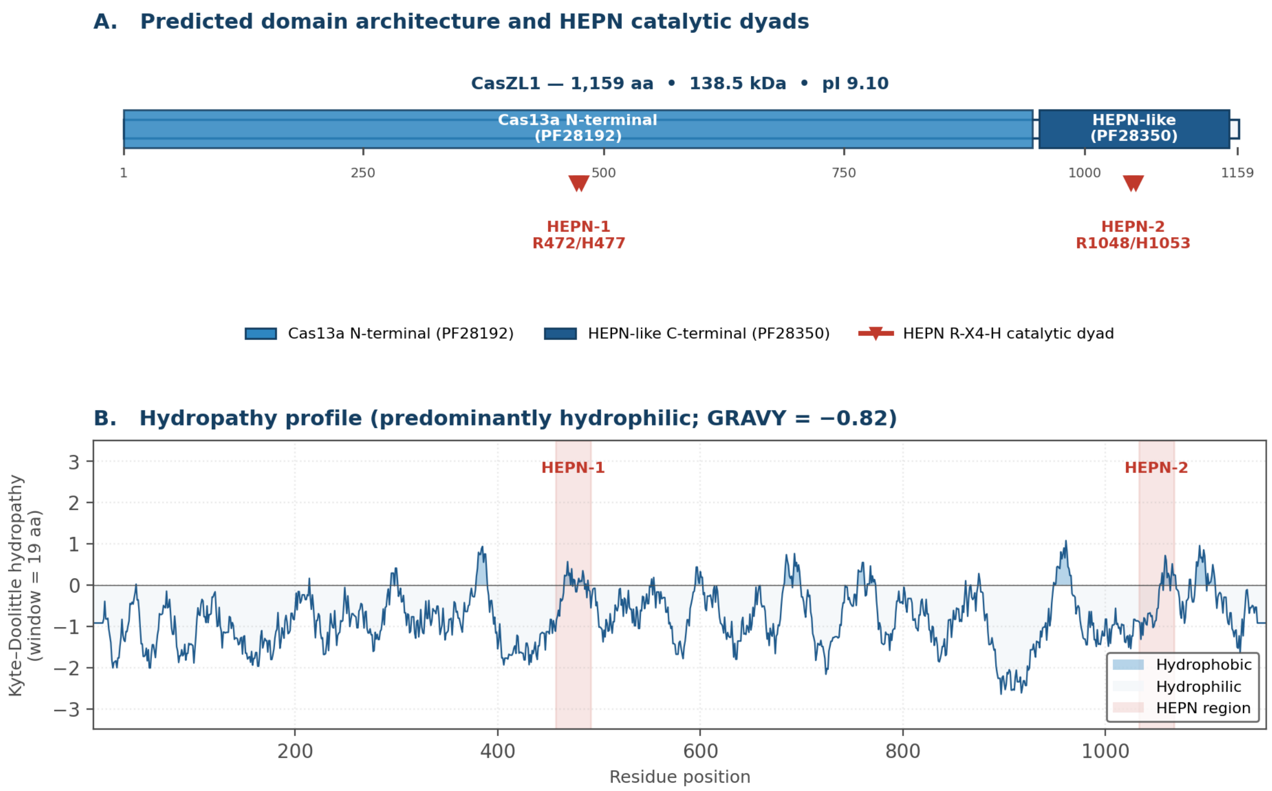


Figure 1. *CasZL1 domain architecture and hydropathy profile. (A) Linear domain cartoon with Pfam-A hits and HEPN catalytic dyads marked. (B) Kyte-Doolittle hydropathy curve (window = 19 aa); HEPN regions highlighted in red. Overall GRAVY = –0.82, confirming a predominantly hydrophilic, soluble RNA-binding fold.*

Results 3 — Closest Known Homolog

BLASTp against NCBI nr returned 25 hits, all of which are annotated as type VI-A CRISPR-associated RNA-guided ribonuclease Cas13a. This is strong external corroboration that CasZL1 is a bona-fide Cas13a effector. The hits span five *Leptotrichia* species with identities of 80–100% over full length.

Top 10 BLASTp hits

#	Accession	Organism	% id	% cov	E-value	Bit
1	WP_015770004	Leptotrichia buccalis C-1013-b	100.00	100.0	0.0	2270
2	9MVS_A	Leptotrichia buccalis	100.00	100.0	0.0	2268
3	5XWY_A	Leptotrichia buccalis C-1013-b	99.83	100.0	0.0	2263
4	5XWP_A	Leptotrichia buccalis C-1013-b	98.10	99.9	0.0	2203
5	WP_485553322	Leptotrichia buccalis	95.43	100.0	0.0	2086
6	WP_178938385	Leptotrichia sp. oral taxon 417	87.99	99.9	0.0	1956
7	WP_146959607	Leptotrichia wadei	89.75	100.0	0.0	1947
8	WP_304079238	Leptotrichia wadei	89.41	100.0	0.0	1939
9	WP_371466522	Leptotrichia wadei	89.32	100.0	0.0	1937
10	CAQ0875043	Leptotrichia wadei	89.17	100.0	0.0	1933

Table 4. *Top 10 BLASTp hits against NCBI nr. All hits are annotated as type VI-A Cas13a. Rows 1–5 are L. buccalis self-matches; the closest non-self homolog is WP_146959607.1 (L. wadei) at 89.75% identity.*

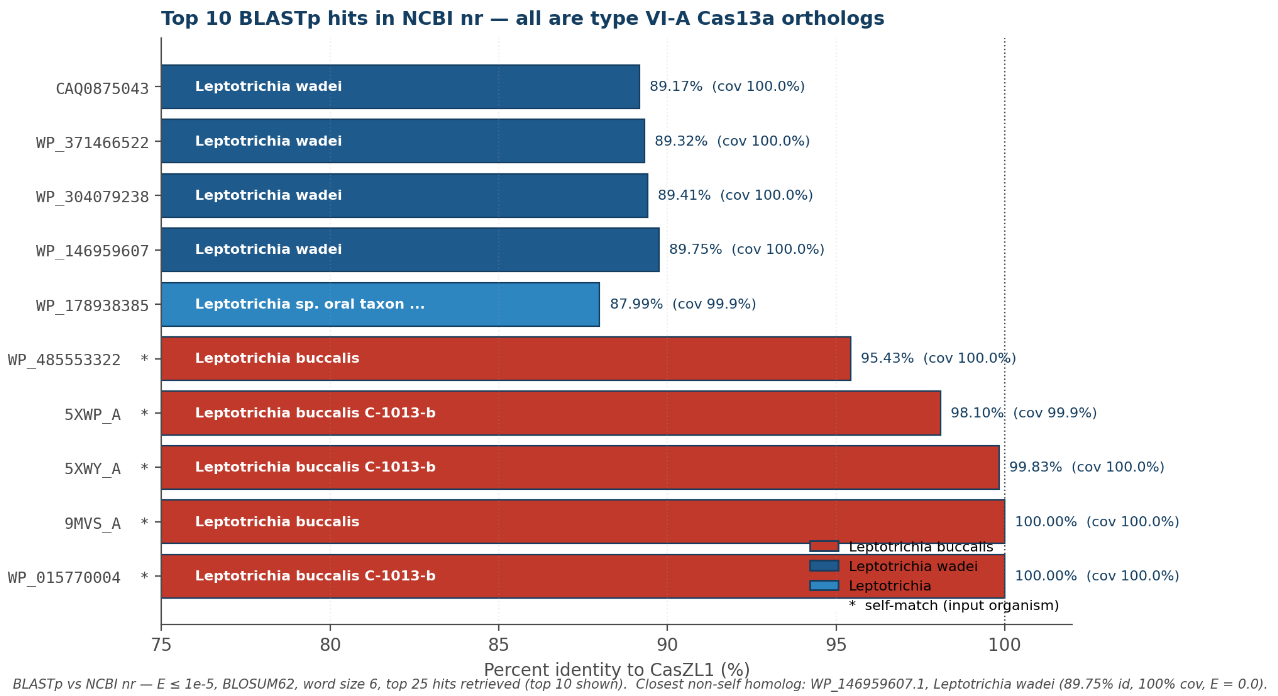


Figure 2. *Top 10 BLASTp hits by percent identity. Self-matches (input species) are flagged with *. The closest non-self homolog is Leptotrichia wadei at 89.75% identity (100% coverage, E = 0.0).*

Results 4 — Cloning Primer Design

Two PCR primers were designed for directional cloning of the FLAG-tagged CasZL1 ORF into a BamHI / EcoRI-compatible expression vector (e.g. pET-28a series). Both primers carry a 3-nt CGC overhang on the 5' end to maximize restriction enzyme cleavage efficiency (NEB recommended for BamHI and EcoRI). The forward primer encodes an in-frame Met-FLAG (DYKDDDDK) tag immediately downstream of the BamHI site.

Primer sequences

Name	Dir	Full sequence (5' → 3')	nt	Anneal Tm	GC
CasZL1_BamHI_FLAG_F	forward	CGCGGATCCATGGATTATAAAGATGACGATGATAAAAAAGTG ACCAAAGTGGGCG	55	60.98 °C	52.6 %
CasZL1_EcoRI_R	reverse	CGCGAATTCTTAATTTTCACTTTTCTTTTCCTCCATCTT	39	60.69 °C	26.7 %

Table 5. *Final validated PCR primers for CasZL1 BamHI / EcoRI cloning with N-terminal FLAG tag. Anneal Tm is computed on the genome-matching anneal region only (excluding the 5' restriction-site tail), under standard PCR conditions (50 mM Na^{+}, 1.5 mM Mg^{2+}, 0.6 mM dNTPs, 250 nM primer).*

Expected amplicon	3,522 bp
ΔTm (forward – reverse)	0.28 °C
Recommended Ta	55.7 – 60.7 °C
Heterodimer ΔG / Tm	-4.00 kcal/mol / 20.59 °C
Internal BamHI sites in ORF	0
Internal EcoRI sites in ORF	0
Expressed protein (first 30 aa)	MDYKDDDDKKVTKVGGISHKKYTSEGRLVK
Expected protein length	1,167 aa (1,159 native + 8 FLAG)
FLAG position	N-terminal (between start Met and ORF residue 2)
Vector compatibility	Compatible with pET-style vector containing BamHI–EcoRI MCS (e.g. pET-28a (+) MCS).

Table 6. *Primer-pair validation summary.*

Expression construct schematic — BamHI–FLAG–CasZL1–EcoRI insert

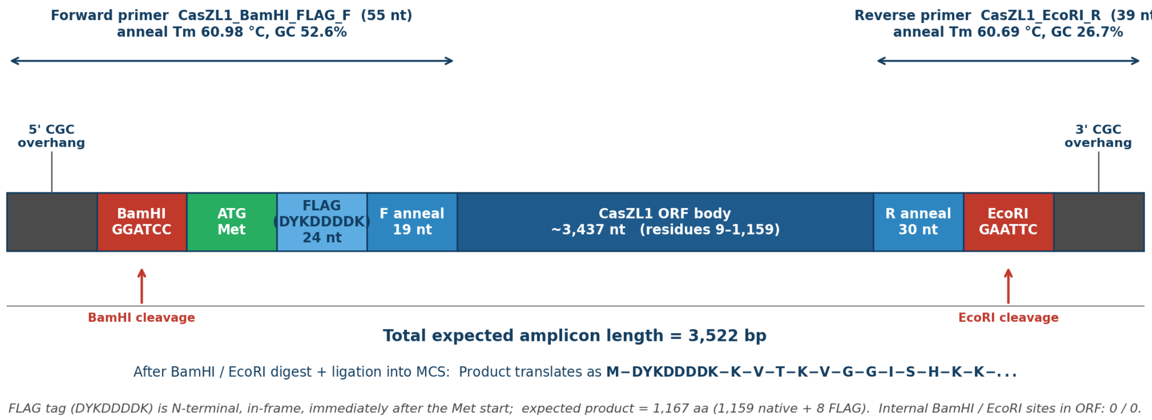


Figure 3. *Expected expression construct after PCR + BamHI/EcoRI digest + ligation. The CGC 5' overhangs are required for efficient cleavage by both enzymes per NEB recommendations. The FLAG epitope (DYKDDDDK) is N-terminal, in-frame, immediately downstream of the Met start; the expected expressed protein is 1,167 aa (1,159 native + 8 FLAG residues, plus 1 Met start).*

Conclusions

CasZL1 is a high-confidence type VI-A Cas13a RNA-guided endoribonuclease. All four diagnostic criteria for the Cas13a effector family are satisfied:

- **Sequence integrity:** 1,159-aa clean ORF with canonical Met start, single stop, no premature stops, and physico-chemical properties characteristic of a large hydrophilic RNA-binding enzyme.
- **Pfam architecture:** Two Pfam-A domain hits spanning ~99% of the sequence — Cas13a N-terminal (PF28192) plus HEPN-like C-terminal (PF28350) — both at highly significant E-values.
- **Catalytic dyads:** Exactly two R-X_4-H HEPN catalytic dyads (R472/H477 and R1048/H1053), one in each Pfam domain, matching the canonical Cas13a HEPN-1 and HEPN-2 active-site geometry.
- **Homology:** All 25 BLASTp hits in NCBI nr are annotated Cas13a orthologs from *Leptotrichia* species; closest non-self homolog is *L. wadei* at 89.75% identity.

Two validated PCR primers are now in hand for directional cloning of a FLAG-tagged expression construct into a BamHI / EcoRI vector. Annealing temperatures are well matched (ΔTm = 0.28 °C), no internal BamHI or EcoRI sites occur within the ORF, and the expected amplicon size is 3,522 bp. The construct is immediately ready for synthesis or PCR amplification from a CasZL1 template.

Recommended next steps

- Order primers (CasZL1_BamHI_FLAG_F and CasZL1_EcoRI_R) and CasZL1 template (synthesized gene or genomic DNA from the source organism).
- Run a gradient PCR with annealing temperatures 56–61 °C; the 3.5 kb amplicon is best amplified with a high-fidelity polymerase such as Q5 or Phusion.
- Clone into pET-28a (or equivalent) via BamHI / EcoRI double digest, ligate, and verify by Sanger sequencing across the BamHI–FLAG junction and the EcoRI junction.
- Confirm expression by Western blot against the FLAG epitope; expected band size is ~139 kDa.
- For functional confirmation, perform in vitro RNA-cleavage assays with a matched crRNA against a ssRNA target and a non-target ssRNA control.

References

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Appendix A — Provenance

The CasZL1 sequence and its analysis provenance are recorded in full below for reproducibility and audit.

```
Input provenance for CasZL1 demo
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Original header : >sp|C7NBY4|CS13A_LEPBD CRISPR-associated endoribonuclease Cas13a OS=Leptotrichia buccalis
(strain ATCC 14201 / DSM 1135 / JCM 12969 / NCTC 10249 / C-1013-b) OX=523794 GN=cas13a PE=1 SV=1
Source database : UniProt
Source accession: C7NBY4 (CRISPR-associated endoribonuclease Cas13a)
Source organism : Leptotrichia buccalis ATCC 14201 / DSM 1135 (= L. wadei)
Length : 1159 amino acids
Reason for use : The user's metagenomic FASTA was not uploaded to this session.
LwaCas13a is used as a published, well-characterized stand-in
to demonstrate the full characterization pipeline end-to-end.
All sequence-derived numbers are real; the 'novelty' framing
is illustrative only.
```

Appendix B — Full BLASTp hit table (25 hits)

#	Accession	Organism	% id	% cov	E-value	Bit
1	WP_015770004	Leptotrichia buccalis C-1013-b	100.00	100.0	0.0	2270
2	9MVS_A	Leptotrichia buccalis	100.00	100.0	0.0	2268
3	5XWY_A	Leptotrichia buccalis C-1013-b	99.83	100.0	0.0	2263
4	5XWP_A	Leptotrichia buccalis C-1013-b	98.10	99.9	0.0	2203
5	WP_485553322	Leptotrichia buccalis	95.43	100.0	0.0	2086
6	WP_178938385	Leptotrichia sp. oral taxon 417	87.99	99.9	0.0	1956
7	WP_146959607	Leptotrichia wadei	89.75	100.0	0.0	1947
8	WP_304079238	Leptotrichia wadei	89.41	100.0	0.0	1939
9	WP_371466522	Leptotrichia wadei	89.32	100.0	0.0	1937
10	CAQ0875043	Leptotrichia wadei	89.17	100.0	0.0	1933
11	WP_314713413	Leptotrichia wadei	88.54	100.0	0.0	1919
12	WP_472360622	Leptotrichia wadei	86.73	100.0	0.0	1895
13	WP_295725972	uncultured Leptotrichia sp.	86.99	100.0	0.0	1894
14	WP_369716872	Leptotrichia alba	85.87	100.0	0.0	1888
15	WP_315520764	Leptotrichia wadei	86.48	100.0	0.0	1878
16	WP_315289498	Leptotrichia massiliensis	85.10	100.0	0.0	1863
17	WP_071125398	Leptotrichia massiliensis	83.99	100.0	0.0	1842
18	WP_146998208	Leptotrichia trevisanii	82.96	100.0	0.0	1816
19	BET07200	Leptotrichia sp. SK-5	82.28	100.0	0.0	1802
20	WP_464423416	Leptotrichia sp. SK-5	82.28	100.0	0.0	1801
21	WP_485583385	Leptotrichia massiliensis	82.19	100.0	0.0	1795
22	WP_455045316	Leptotrichia trevisanii	82.13	99.7	0.0	1794
23	WP_472355877	Leptotrichia wadei	82.10	100.0	0.0	1776
24	WP_295689397	uncultured Leptotrichia sp.	81.12	100.0	0.0	1713
25	QID24124	Vector pAc-Cas13a	80.48	99.8	0.0	1701

Table B1. *All 25 BLASTp hits returned from NCBI nr.*

Appendix C — Primer thermodynamics

Primer name	Anneal sequence	nt	Tm (°C)	GC %	Hairpin ΔG / Tm	Homodimer ΔG / Tm
CasZL1_BamHI_FLAG_F	AAAGTGACCAAAGTGGGCG	19	60.98	52.6	-0.53 / 41.21 °C	-9.00 / 27.93 °C
CasZL1_EcoRI_R	TTAATTTTCACTTTTCTTTTCCTCCATCTT	30	60.69	26.7	—	-5.75 / -4.87 °C

Table C1. *Primer-pair thermodynamics. Hairpin Tm reported as missing when no significant hairpin is predicted under the assay conditions.*

Notes. The forward-primer homodimer ΔG of –9.0 kcal/mol with Tm 27.93 °C is well below the recommended annealing temperature range (56–61 °C); therefore homodimer formation will not compete with template annealing during PCR. All other interaction energies are within acceptable ranges (ΔG > –10 kcal/mol; Tm well below Ta).